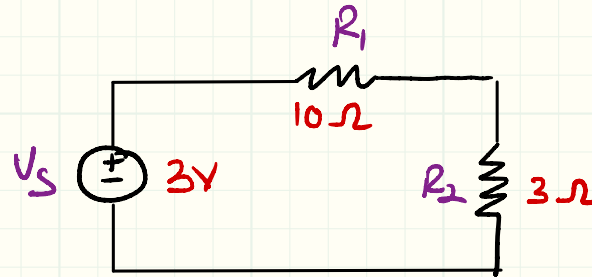
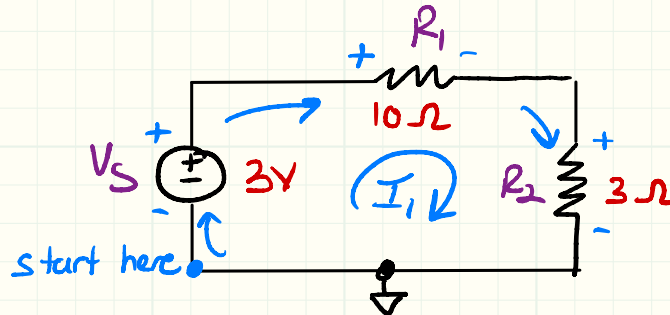


Problem ①

Using mesh Analysis, solve the currents @ each mesh of the following circuit.



Step ①: Label the circuit:



One Loop
→ one mesh
→ mesh current I_1

Step ②: Set up KVL Eq @ meshes:

KVL @ mesh ① → Sum of Voltages around mesh = zero

$$-V_S + R_1(I_1) + R_2(I_1) = 0 \quad - \text{Eq ①}$$

Step ③: Plug In Given Values:

$$-V_S + R_1(I_1) + R_2(I_1) = 0 \quad - \text{Eq ①}$$

↓

$$-3 + 10I_1 + 3I_1 = 0 \quad - \text{Eq ①}$$

Step ④: Rewrite for System of Eq Solve:

$$-3 + 10I_1 + 3I_1 = 0$$

↓

$$10I_1 + 3I_1 = 3 \quad \& \text{ Go ahead + solve } \rightarrow \text{ only } \underline{\text{one unknown!}}$$

↓

$$13I_1 = 3$$

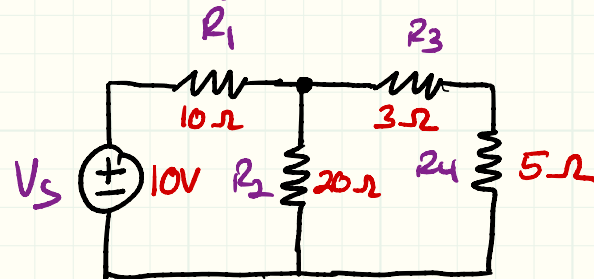
↓

$$I_1 = \frac{3}{13}$$

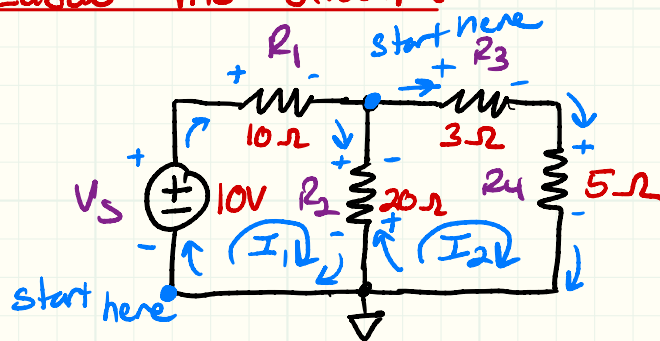
$$I_1 = \frac{3}{13} \text{ Amps}$$

Problem ②

Using Mesh Analysis, solve the currents @ each mesh of the following circuit.



Step ①: Label the circuit:



Two Loops
Two Mesh
Mesh Currents
 I_1 and I_2

Step ②: Set up KVL Eq @ meshes:

KVL @ Mesh ① $\rightarrow$ Sum of Voltages around mesh = zero

$$-V_s + R_1(I_1) + R_2(I_1 - I_2) = 0 \quad - \text{Eq ①}$$

KVL @ Mesh ② $\rightarrow$ Sum of Voltages around mesh = zero

$$R_3(I_2) + R_4(I_2) + R_2(I_2 - I_1) = 0 \quad - \text{Eq ②}$$

Step ③: Plug In Given Values:

$$-V_s + R_1(I_1) + R_2(I_1 - I_2) = 0 \quad - \text{Eq ①}$$

$\downarrow$

$$-10 + 10I_1 + 20(I_1 - I_2) = 0 \quad - \text{Eq ①}$$

$$R_3(I_2) + R_4(I_2) + R_2(I_2 - I_1) = 0 \quad - \text{Eq ②}$$

$\downarrow$

$$3I_2 + 5I_2 + 20(I_2 - I_1) = 0 \quad - \text{Eq ②}$$

Step ④: Rewrite for System of Eq Solve:

$$-10 + 10I_1 + 20(I_1 - I_2) = 0 \quad - \text{Eq ①}$$

$\downarrow$

$$10I_1 + 20I_1 - 20I_2 = 10$$

$\downarrow$

$$30I_1 - 20I_2 = 10 \quad - \text{Eq ③}$$

Problem ②

$$3I_2 + 5I_2 + 20(I_2 - I_1) = 0 \quad - \text{Eq ②}$$

$$\downarrow 3I_2 + 5I_2 + 20I_2 - 20I_1 = 0$$

$$\downarrow 28I_2 - 20I_1 = 0$$

$$\downarrow -20I_1 + 28I_2 = 0 \quad - \text{Eq ④}$$

Step ⑤: Solve the System of Eq

$$30I_1 - 20I_2 = 10 \quad - \text{Eq ③}$$

$$-20I_1 + 28I_2 = 0 \quad - \text{Eq ④}$$

$$A = \begin{bmatrix} 30 & -20 \\ -20 & 28 \end{bmatrix} ; B = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

$$Y = A^{-1} * B = \begin{bmatrix} 30 & -20 \\ -20 & 28 \end{bmatrix}^{-1} * \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0.636 \\ 0.454 \end{bmatrix}$$

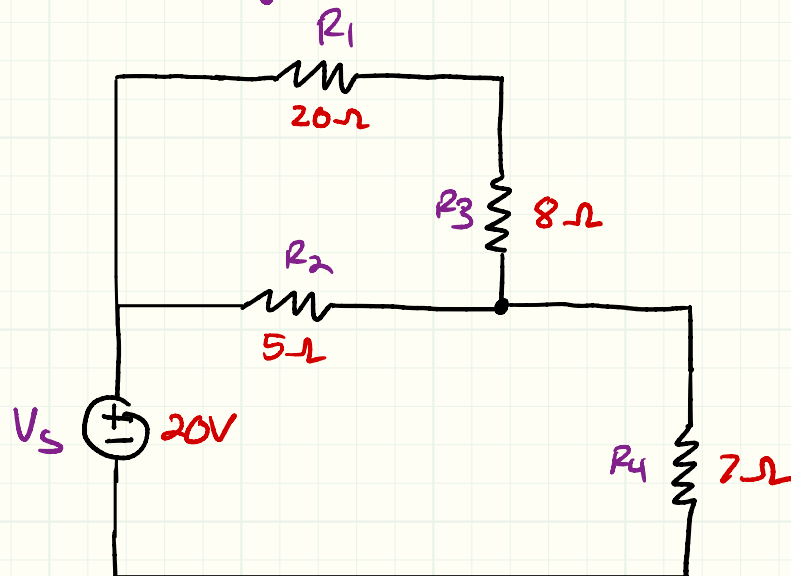
Remember $\rightarrow Y = \text{Var Answer}$
 $Y = \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$

$$I_1 = 0.636 \text{ Amps}$$

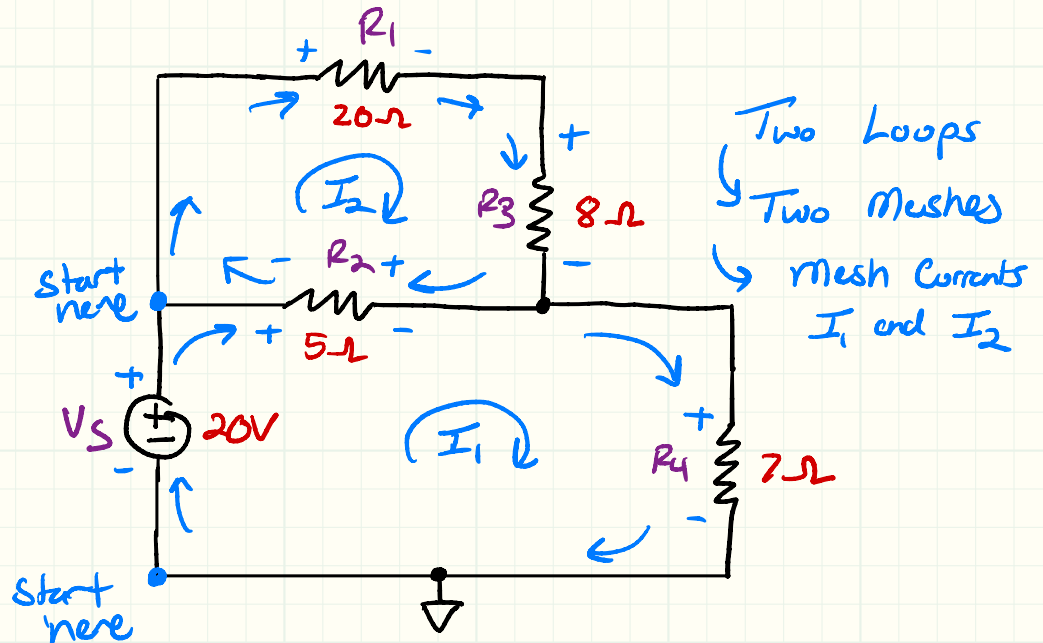
$$I_2 = 0.454 \text{ Amps}$$

Problem ③

Using mesh Analysis, solve the currents @ each mesh of the following circuit.



Step ①: Label the circuit:



Step ②: Set up KVL Eq @ meshes:

KVL @ Mesh ① $\rightarrow$ Sum of Voltages around mesh = zero

$$-V_S + R_2(I_1 - I_2) + R_4 I_1 = 0 \quad - \text{Eq ①}$$

KVL @ Mesh ② $\rightarrow$ Sum of Voltages around mesh = zero

$$R_1(I_2) + R_3(I_2) + R_2(I_2 - I_1) = 0 \quad - \text{Eq ②}$$

Problem ③Step ③: Plug In Given Values:

$$-V_s + R_2(I_1 - I_2) + R_4 I_1 = 0 \quad - \text{Eq ①}$$

$$\downarrow$$
$$-20 + 5(I_1 - I_2) + 7I_1 = 0 \quad - \text{Eq ①}$$

$$R_1(I_2) + R_3(I_2) + R_2(I_2 - I_1) = 0 \quad - \text{Eq ②}$$

$$\downarrow$$
$$20I_2 + 8I_2 + 5(I_2 - I_1) = 0 \quad - \text{Eq ②}$$

Step ④: Rewrite for System of Eq Solve:

$$-20 + 5(I_1 - I_2) + 7I_1 = 0$$

$$\downarrow$$
$$5I_1 - 5I_2 + 7I_1 = 20$$

$$\downarrow$$
$$12I_1 - 5I_2 = 20 \quad - \text{Eq ③}$$

$$20I_2 + 8I_2 + 5(I_2 - I_1) = 0$$

$$\downarrow$$
$$20I_2 + 8I_2 + 5I_2 - 5I_1 = 0$$

$$\downarrow$$
$$33I_2 - 5I_1 = 0$$

$$\downarrow$$
$$-5I_1 + 33I_2 = 0 \quad - \text{Eq ④}$$

Step ⑤: Solve the System of Eq

$$12I_1 - 5I_2 = 20 \quad - \text{Eq ③}$$

$$-5I_1 + 33I_2 = 0 \quad - \text{Eq ④}$$

$$Y = \begin{bmatrix} 12 & -5 \\ -5 & 33 \end{bmatrix}^{-1} * \begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

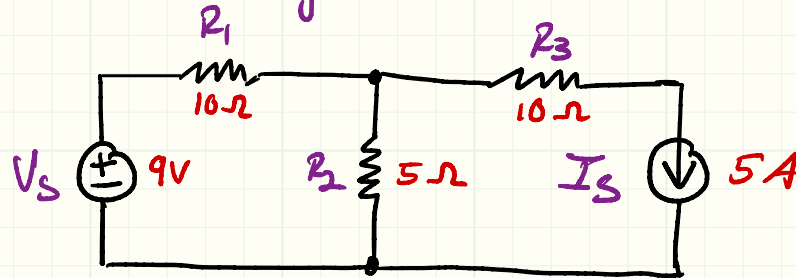
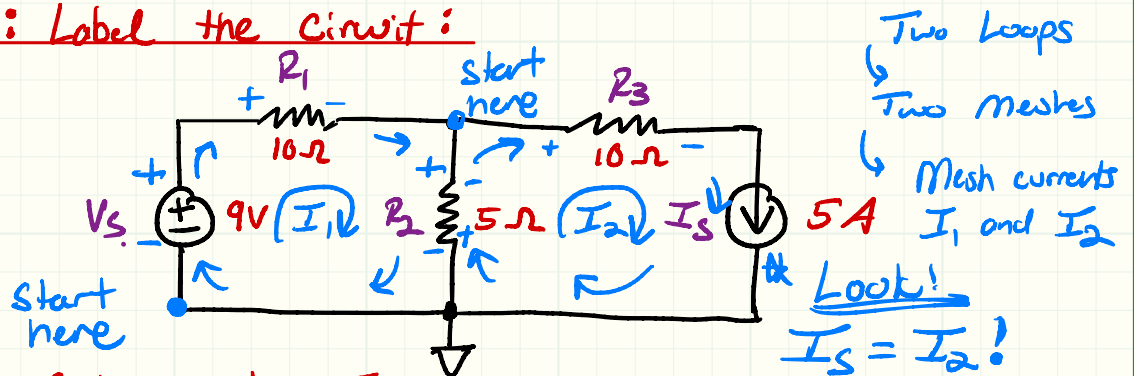
$$Y = \begin{bmatrix} 1.779 \\ 0.270 \end{bmatrix}$$

$$I_1 = 1.779 \text{ Amps}$$

$$I_2 = 0.270 \text{ Amps}$$

Problem ④

Using Mesh Analysis, solve the currents @ each mesh of the following circuit.

Step ①: Label the circuit:Step ②: Set up KVL Eq @ meshes: Given Solution Pt. 2!

KVL @ Mesh ① → Sum of voltages around mesh = zero

$$-V_s + R_1(I_1) + R_2(I_1 - I_2) = 0 \quad - \text{Eq ①}$$

KVL @ Mesh ② → Sum of voltages around mesh = zero

$$\text{Given that } \underline{I_2 = 5 \text{ Amps!}} \quad - \text{Eq ②}$$

Step ③: Plug In Given Values:

$$-V_s + R_1(I_1) + R_2(I_1 - I_2) = 0 \quad - \text{Eq ①}$$

$$\downarrow -9 + 10I_1 + 5(I_1 - I_2) = 0 \quad - \text{Eq ②}$$

* Since we were given $I_2 \rightarrow$ Just Solve for I_1 !

$$\downarrow -9 + 10I_1 + 5I_1 - 5(5) = 0$$

$$15I_1 = 34$$

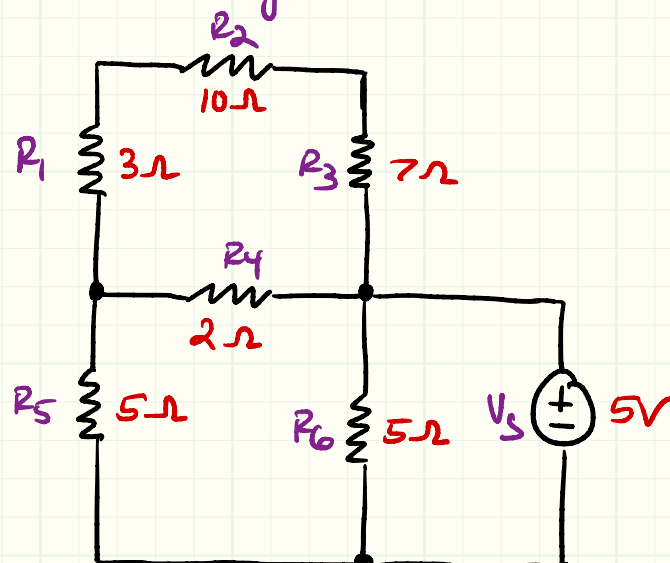
$$I_1 = \frac{34}{15}$$

$$I_1 = 2.267 \text{ Amps}$$

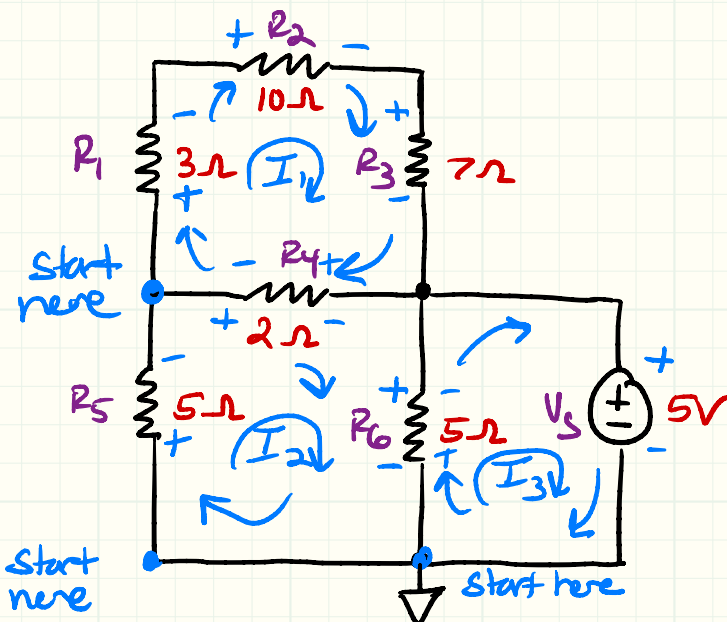
$$I_2 = 5 \text{ Amps}$$

Problem ⑤

Using Mesh Analysis, solve the currents @ each mesh of the following circuit.



Step ①: Label the circuit:



Step ②: Set up KVL eq @ meshes:

KVL @ Mesh ① → Sum of Voltages around mesh = zero

$$R_1(I_1) + R_2(I_1) + R_3(I_1) + R_4(I_1 - I_2) = 0 \quad - \text{Eq ①}$$

KVL @ Mesh ② → Sum of Voltages around mesh = zero

$$R_5(I_2) + R_4(I_2 - I_1) + R_6(I_2 - I_3) = 0 \quad - \text{Eq ②}$$

KVL @ Mesh ③ → Sum of Voltages around mesh = zero

$$R_6(I_3 - I_2) + 5 = 0 \quad - \text{Eq ③}$$

Problem ⑤Step ③: Plug In Given Values:

$$R_1(I_1) + R_2(I_1) + R_3(I_1) + R_4(I_1 - I_2) = 0 \quad - E_a \quad ①$$

$$\downarrow$$

$$3I_1 + 10I_1 + 7I_1 + 2(I_1 - I_2) = 0 \quad - E_a \quad ①$$

$$R_5(I_2) + R_4(I_2 - I_1) + R_6(I_2 - I_3) = 0 \quad - E_a \quad ②$$

$$\downarrow$$

$$5I_2 + 2(I_2 - I_1) + 5(I_2 - I_3) = 0 \quad - E_a \quad ②$$

$$R_6(I_3 - I_2) + 5 = 0 \quad - E_a \quad ③$$

$$\downarrow$$

$$5(I_3 - I_2) + 5 = 0 \quad - E_a \quad ③$$

Step ④: Rewrite for System of E_a Solve:

$$3I_1 + 10I_1 + 7I_1 + 2(I_1 - I_2) = 0$$

$$\downarrow$$

$$20I_1 + 2I_1 - 2I_2 = 0$$

$$\downarrow$$

$$22I_1 - 2I_2 = 0 \quad - E_a \quad ④$$

$$5I_2 + 2(I_2 - I_1) + 5(I_2 - I_3) = 0$$

$$\downarrow$$

$$5I_2 + 2I_2 - 2I_1 + 5I_2 - 5I_3 = 0$$

$$\downarrow$$

$$12I_2 - 2I_1 - 5I_3 = 0$$

$$\downarrow$$

$$-2I_1 + 12I_2 - 5I_3 = 0 \quad - E_a \quad ⑤$$

$$5(I_3 - I_2) + 5 = 0$$

$$\downarrow$$

$$5I_3 - 5I_2 + 5 = 0$$

$$\downarrow$$

$$0I_1 - 5I_2 + 5I_3 = -5 \quad - E_a \quad ⑥$$

Step ⑤: Solve the System of E_a

$$22I_1 - 2I_2 + 0I_3 = 0 \quad - E_a \quad ④$$

$$-2I_1 + 12I_2 - 5I_3 = 0 \quad - E_a \quad ⑤$$

$$0I_1 - I_2 + 5I_3 = -5 \quad - E_a \quad ⑥$$

$$Y = \begin{bmatrix} 22 & -2 & 0 \\ -2 & 12 & -5 \\ 0 & -5 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ -5 \end{bmatrix}$$

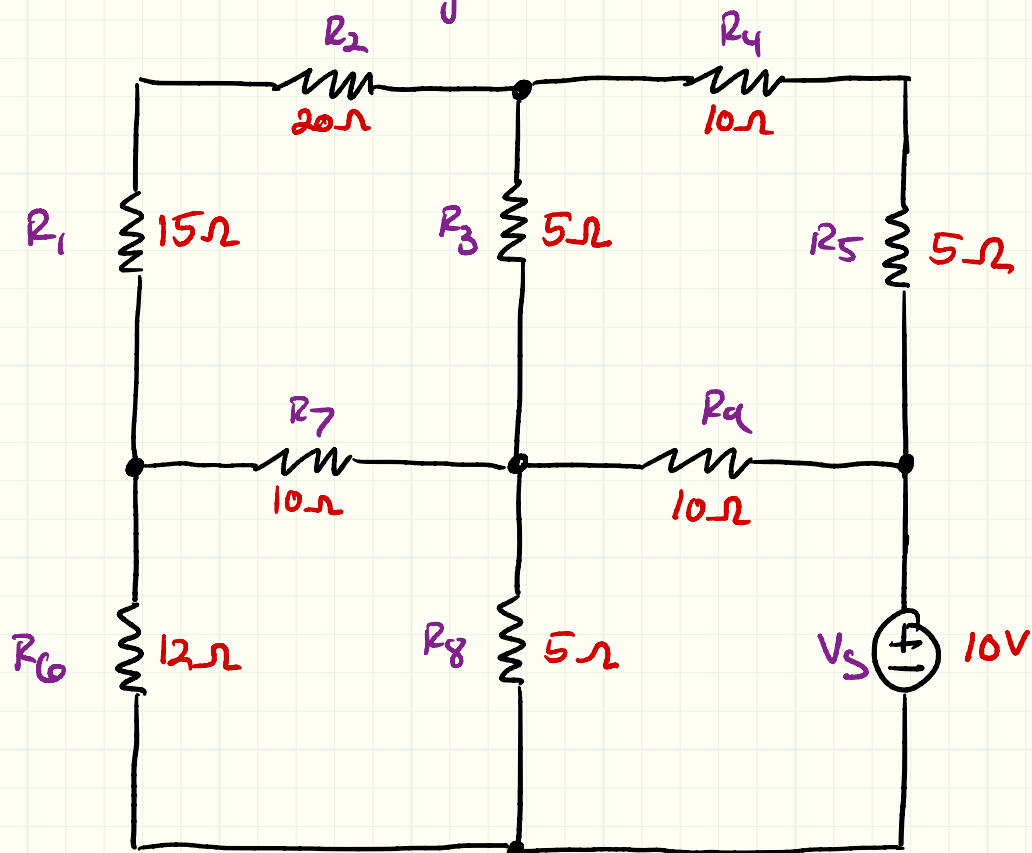
$$I_1 = -0.066 \text{ Amps}$$

$$I_2 = -0.733 \text{ Amps}$$

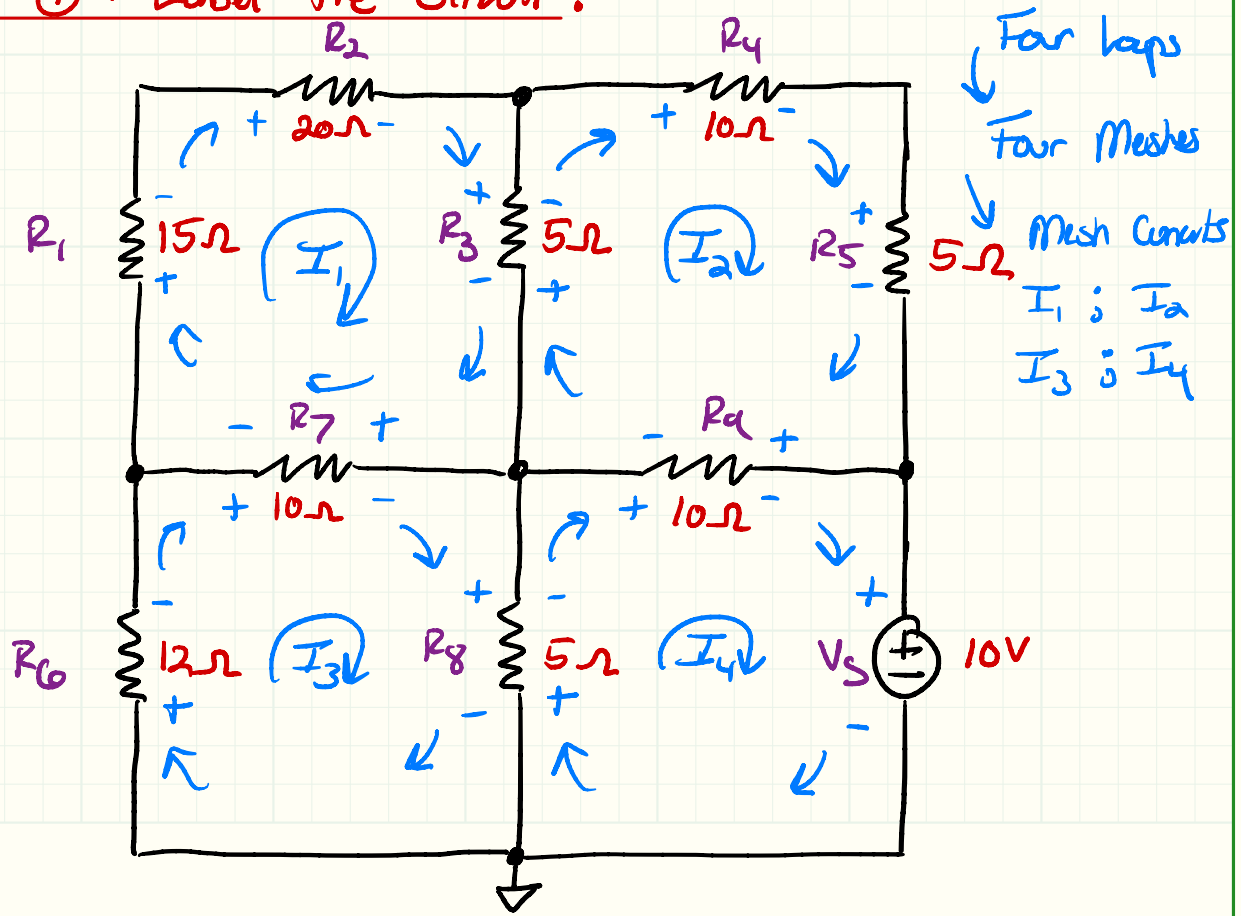
$$I_3 = -1.733 \text{ Amps}$$

Problem (6)

Using Mesh Analysis, solve the currents @ each mesh of the following circuit.



Step (1): Label the circuit:



Problem ⑥Step ②: Set up KVL Eq @ meshes:

$$\text{KVL @ Mesh ①} \rightarrow \text{Sum of Voltages around mesh} = \text{zero}$$

$$R_1(I_1) + R_2(I_1) + R_3(I_1 - I_2) + R_7(I_1 - I_3) = 0 - E_a \quad \text{①}$$

$$\text{KVL @ Mesh ②} \rightarrow \text{Sum of Voltages around mesh} = \text{zero}$$

$$R_3(I_2 - I_1) + R_4(I_2) + R_5(I_2) + R_9(I_2 - I_4) = 0 - E_a \quad \text{②}$$

$$\text{KVL @ Mesh ③} \rightarrow \text{Sum of Voltages around mesh} = \text{zero}$$

$$R_6(I_3) + R_7(I_3 - I_1) + R_8(I_3 - I_4) = 0 - E_a \quad \text{③}$$

$$\text{KVL @ Mesh ④} \rightarrow \text{Sum of Voltages around mesh} = \text{zero}$$

$$R_8(I_4 - I_3) + R_9(I_4 - I_2) + V_S = 0 - E_a \quad \text{④}$$

Step ③: Plug In Given Values:

$$R_1(I_1) + R_2(I_1) + R_3(I_1 - I_2) + R_7(I_1 - I_3) = 0 - E_a \quad \text{①}$$

$$\downarrow$$

$$15I_1 + 20I_1 + 5(I_1 - I_2) + 10(I_1 - I_3) = 0 - E_a \quad \text{①}$$

$$R_3(I_2 - I_1) + R_4(I_2) + R_5(I_2) + R_9(I_2 - I_4) = 0 - E_a \quad \text{②}$$

$$\downarrow$$

$$5(I_2 - I_1) + 10I_2 + 5I_2 + 10(I_2 - I_4) = 0 - E_a \quad \text{②}$$

$$R_6(I_3) + R_7(I_3 - I_1) + R_8(I_3 - I_4) = 0 - E_a \quad \text{③}$$

$$\downarrow$$

$$12I_3 + 10(I_3 - I_1) + 5(I_3 - I_4) = 0 - E_a \quad \text{③}$$

$$R_8(I_4 - I_3) + R_9(I_4 - I_2) + V_S = 0 - E_a \quad \text{④}$$

$$\downarrow$$

$$5(I_4 - I_3) + 10(I_4 - I_2) + 10 = 0 - E_a \quad \text{④}$$

Step ④: Rewrite for Sys of Eq Solve

$$15I_1 + 20I_1 + 5(I_1 - I_2) + 10(I_1 - I_3) = 0$$

$$\downarrow$$

$$35I_1 + 5I_1 - 5I_2 + 10I_1 - 10I_3 = 0$$

$$\downarrow$$

$$50I_1 - 5I_2 - 10I_3 + 0I_4 = 0 - E_a \quad \text{⑤}$$

Problem ⑥

$$5(I_2 - I_1) + 10I_2 + 5I_2 + 10(I_2 - I_4) = 0$$

$$\downarrow$$

$$5I_2 - 5I_1 + 15I_2 + 10I_2 - 10I_4 = 0$$

$$\downarrow$$

$$-5I_1 + 30I_2 + 0I_3 - 10I_4 = 0 \quad \text{--- Eq ⑥}$$

$$12I_3 + 10(I_3 - I_1) + 5(I_3 - I_4) = 0$$

$$\downarrow$$

$$12I_3 + 10I_3 - 10I_1 + 5I_3 - 5I_4 = 0$$

$$\downarrow$$

$$-10I_1 + 0I_2 + 27I_3 - 5I_4 = 0 \quad \text{--- Eq ⑦}$$

$$5(I_4 - I_3) + 10(I_4 - I_2) + 10 = 0$$

$$\downarrow$$

$$5I_4 - 5I_3 + 10I_4 - 10I_2 + 10 = 0$$

$$\downarrow$$

$$0I_1 - 10I_2 - 5I_3 + 15I_4 = -10 \quad \text{--- Eq ⑧}$$

Step ⑤: Sys of Eq Solve

$$Y = \begin{bmatrix} 50 & -5 & -10 & 0 \\ -5 & 30 & 0 & -10 \\ -10 & 0 & 27 & -5 \\ 0 & -10 & -5 & 15 \end{bmatrix}^{-1} * \begin{bmatrix} 0 \\ 0 \\ 0 \\ -10 \end{bmatrix}$$

$$Y = \begin{bmatrix} -0.073 \\ -0.330 \\ -0.204 \\ -0.955 \end{bmatrix}$$

$$I_1 = -0.073 \text{ Amps}$$

$$I_2 = -0.330 \text{ Amps}$$

$$I_3 = -0.204 \text{ Amps}$$

$$I_4 = -0.955 \text{ Amps}$$